

Where Should Machine Learning Go in Hybrid Trading Systems?

Matched-Budget Component Attribution Across Three Markets

Version 4 (refined, with open-source reference implementation) — code, data, and tests:
https://github.com/Cannonbold2412/ML_Component_Attribution

Abstract

Hybrid trading systems that combine rule-based signal generation with machine-learned components are common in practice, but the literature rarely asks a decomposable question: at which specific decision point does an ML component actually help, and where does it hurt? We introduce **Matched-Budget Component Attribution (MBCA)**, a general-purpose methodology for answering that question in any hybrid decision system: hold a validated rule-based control fixed, replace exactly one insertion point at a time with a matched-budget machine-learned counterpart — identical features, identical model class, identical evaluation protocol — and report every resulting configuration's forward-tested statistical significance, not only the configuration that happened to look best. We apply MBCA to one already-validated rule-based trading strategy — a JMA fast/slow crossover with an ATR stop/target (or trailing) exit stack, validated with a strict freeze → holdout → forward-tail protocol on three markets (forex, commodities, Indian equities) — across five insertion points that span a natural component taxonomy: an independent bar-level gate (regime filter); two narrowing components that filter or re-weight decisions the rule-based system already makes (meta-labeling, position sizing); and two replacing components that substitute the rule-based mechanism outright (entry generation, exit policy). Every component shares the same 8 backward-only features, the same LightGBM configuration, the same freeze discipline, and the same deflated-Sharpe / bootstrap-CI statistical stack.

Across 5 individual components plus 4 combination experiments (Entry+Exit, Entry+Sizing, Exit+Sizing, and an empirically-chosen Best Combined pairing of Meta-Labeling with Position Sizing) × 3 markets × 2 evaluation windows — 60 testable configurations with a bootstrap CI available — **zero show a statistically significant improvement over the rule-based baseline**, while 20 show a statistically significant loss. We summarize the full grid as the **ML Opportunity Matrix**, a component/combination × market decision-support artifact with a mechanically-derived recommendation column, and a companion decision flowchart. The evidence is component-specific and market-specific, not uniform: narrowing components are gentler and occasionally directionally positive, concentrated in Indian equities, while replacing components are more volatile and, for exit specifically, actively harmful — ML Exit inflates trade count 2.7×–4.2× (early learned exits un-block JMA crossovers a longer-held rule-based trade would have absorbed) and loses significantly in 4 of 5 tested configurations. On the original forex data, every ML intervention tried loses (a finding the independent replication does not reproduce). Best Combined — Meta-Labeling filtering trades, then Position Sizing re-weighting the survivors — produces the campaign's single closest-to-significant result: Indian-equities holdout Sharpe 2.84 → 4.01 (bootstrap CI on the per-day Sharpe difference [−0.050,0.176], i.e. roughly [−0.8,2.8] in annualized units) and forward Sharpe 1.05 → 2.70 (CI [−0.087,0.293], roughly [−1.4,4.7] annualized), with a directionally consistent, non-significant improvement in commodities on both windows too. The paper's contribution is MBCA itself, the ML Opportunity Matrix and decision framework it produces, and the resulting map of where ML helps, hurts, or is a wash in one concrete hybrid system — not a claim of a validated ML-enhanced trading strategy, which the evidence here does not support, though Best Combined in Indian equities is the one configuration worth a second-freeze-date replication before ruling out entirely. An independent, open-source re-implementation of the full protocol (**mbca**, released with this paper) reproduces the headline — 0 of 60 significant improvements, ML Exit significantly harmful in every market via the same turnover mechanism, 49 of 60 configurations agreeing in sign — but not the paper's forex result, and only half of Best Combined's near-miss (see the Independent Replication section).

Matched-Budget Component Attribution (MBCA)

Where does ML add value in a hybrid trading system?

	Forex	Commodities	Indian Equities
Regime Filter			
Meta-Labeling			
Entry			
Exit			
Position Sizing			

0 of 60 configurations show a significant gain; narrowing components (meta-labeling, sizing) show mild, unreplicated promise in Indian equities.

Keywords: matched-budget component attribution, hybrid decision systems, component taxonomy, ML opportunity matrix, decision framework, meta-labeling, regime filtering, walk-forward validation, deflated Sharpe ratio, JMA, ATR

Introduction

A large fraction of practitioner and academic “ML for trading” work reports one hybrid configuration — a rule-based signal gated, filtered, or replaced by a learned model — against a baseline, and treats a Sharpe improvement (or lack of one) as evidence about “machine learning in trading” generally. This conflates two very different questions: *does ML help*, and *does ML help at THIS specific decision point, with THIS feature set and model class, holding everything else fixed*. A hybrid pipeline has several natural insertion points — which bars are admissible (regime filter), whether a specific signal-fired trade should be taken (meta-labeling), which bars generate entries at all (entry generation), when to exit (exit policy), and how much capital to risk per trade (position sizing) — and a single aggregate comparison cannot say which of these, if any, is where the reported improvement (or degradation) actually comes from.

We introduce **Matched-Budget Component Attribution (MBCA)** to answer that decomposed question directly and reproducibly. MBCA is a controlled-variable methodology for any hybrid decision system that combines a rule-based control with learned components: hold the data, the feature set, the model class and hyperparameters, the training/freeze discipline, and the evaluation protocol fixed, and vary only *which single insertion point* the ML component replaces. Because every comparison shares an identical budget, a resulting Sharpe delta can be attributed to the insertion point itself rather than to a confound of feature-engineering effort, tuning effort, or dataset choice that would otherwise differ from one paper’s hybrid configuration to the next.

This paper demonstrates MBCA on one already-validated rule-based system: a JMA fast/slow crossover with an ATR-based stop/target (or trailing-stop) exit, tested with a strict freeze → holdout → forward-tail protocol on three markets (a 4-pair forex sleeve, a 5-instrument commodities sleeve, and a 48-stock Indian-equities sleeve with annual sector curation) that this research program had already validated as a standalone rule-based strategy before any ML was introduced. We hold the dataset, features, model class (LightGBM, identical hyperparameters), training procedure, freeze discipline, transaction costs, and evaluation protocol fixed, and replace exactly one component at a time with its ML counterpart.

Research question: *Where, specifically, does replacing one component of an already-validated hybrid decision system with a matched-budget machine-learned counterpart create value — and where does it instead destroy value or do nothing?*

Applying MBCA to this system surfaces a taxonomy of insertion points by how much they can alter the rule-based system’s decisions: *independent* components (a bar-level gate orthogonal to the entry signal), *narrowing* components (filter or re-weight decisions the rule-based

system already makes), and *replacing* components (substitute the rule-based mechanism outright). Across 60 tested configurations, narrowing components show a mild, non-significant, market-concentrated positive pattern; replacing components perform worse, sometimes catastrophically; and the independent regime filter is a dead end everywhere tested. We package this evidence as the **ML Opportunity Matrix** (Figure 2), a compact decision-support artifact, and a companion decision flowchart (Figure 5).

Contributions.

- **MBCA**, a matched-budget, controlled-variable methodology for attributing ML’s contribution to a specific insertion point in a hybrid decision system, reusable beyond this trading strategy and, we argue, beyond trading (Section 8).
- A **component taxonomy** (independent / narrowing / replacing) that predicts, ex ante, which insertion points are gentler or more volatile to convert to ML.
- The **ML Opportunity Matrix** and an accompanying **decision flowchart** — reusable artifacts that turn a 60-row results table into a practical, mechanically-derived recommendation per component per market.
- A full evidence grid across 5 components, 4 combinations, 3 markets, and 2 evaluation windows (60 configurations, none cherry-picked), including the losses.

Research Gap

Ablation studies of ML-augmented trading pipelines exist (e.g. a 2025 cross-sectional A-share turnover pipeline ablates its component modules and attributes annualized-return contribution per module; see Section 3), but they typically ablate *modules of one specific proprietary pipeline architecture*, not *insertion points of a single control mechanism into an already-fixed rule-based strategy* with identical features/model/hyperparameters held across every component. No work we found formalizes this constraint as a named, reusable methodology, runs the same feature set and model class through five distinct insertion points of one rule-based strategy with a uniform freeze/holdout/forward-tail protocol and bootstrap-CI significance testing at every insertion point, and reports the full grid of results — including the (numerous) negative ones — as a decision-support artifact rather than the single configuration that happened to look best. MBCA fills that gap: it is the methodology, not merely the experiment, that is missing from the literature.

Related Work

Meta-labeling. López de Prado’s meta-labeling [1] separates the “which direction” decision (the primary model) from “should I take this specific signal” (a secondary classifier trained on realized outcomes) — exactly the mechanism this paper’s ML Meta-Labeling component implements, though the primary model here is a fixed rule (JMA crossover), not a learned one. In MBCA’s taxonomy (Section 4.2), meta-labeling is a canonical *narrowing* component: it can only reject decisions the primary system already made, never invent new ones.

Regime-conditioned and hybrid rule+ML systems. A 2026 hybrid system integrating technical rules, ML, and financial sentiment for regime-adaptive equity strategies [2] reports aggregate improvements over a rule-only baseline but does not decompose which sub-component is responsible — precisely the conflation MBCA is designed to resolve. A 2025 deep-learning-enhanced, multi-day-turnover A-share pipeline [3] is the closest methodological precedent we found: it ablates its five pipeline modules one at a time and reports each module’s marginal contribution to annualized return — the same spirit as MBCA’s attribution table, but over a different, proprietary module set, without a uniform ML feature/model budget held across modules, without a named, reusable methodology, and without holdout-to-forward-tail generalization testing per module.

Position sizing and volatility-managed portfolios. Moreira and Muir’s volatility-managed portfolios [4] scale exposure by a *realized-volatility* signal at the portfolio level; this paper’s ML Position Sizing component instead scales exposure by a *predicted-win-probability* signal at the individual-trade level — another *narrowing* component in MBCA’s taxonomy, since it re-weights rather than rejects.

Time-series momentum. Moskowitz, Ooi and Pedersen [5] is the standard reference for the rule-based trend-following family this paper’s baseline belongs to; our contribution is orthogonal to theirs.

Gap. No work we found holds a rule-based strategy, its data, its features, its model class, and its evaluation protocol fixed while sweeping which single component ML replaces, names and formalizes that constraint as a reusable methodology, and reports every resulting configuration’s forward-tested significance rather than the best one. MBCA is that methodology; this paper is its first full application.

Methodology

Matched-Budget Component Attribution (MBCA)

Definition. MBCA is a controlled-variable methodology for decomposing a hybrid decision system’s performance by insertion point. It applies whenever a system combines a validated, rule-based control with one or more points

where a learned model could instead make the decision. MBCA holds five things fixed across every tested insertion point:

- the **data** each component’s model is trained and evaluated on;
- the **feature set** available to every learned component;
- the **model class and hyperparameters** (no per-component tuning);
- the **training/freeze discipline** (fit once, never refit on evaluation data); and
- the **evaluation protocol and statistical test** used to call a result significant.

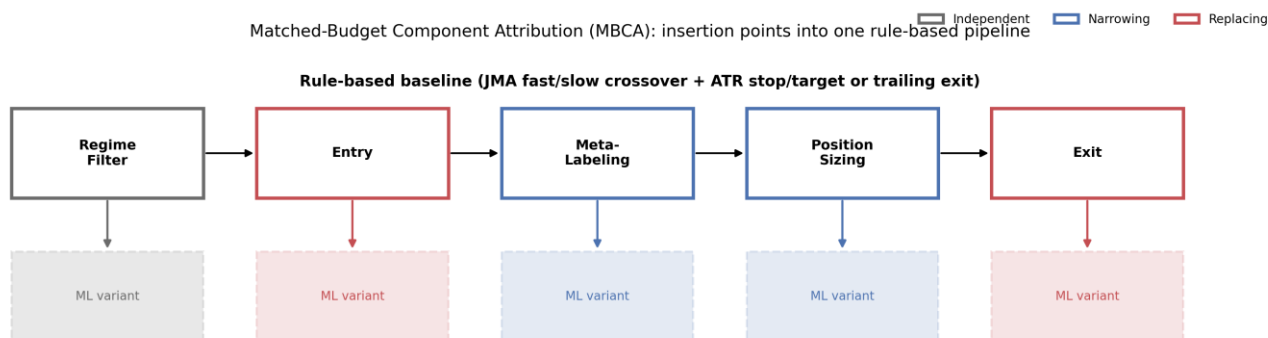
Only the *insertion point* — which single decision in the pipeline the learned model replaces or modifies — is allowed to vary between configurations. This is what makes the resulting Sharpe deltas attributable to the insertion point itself, rather than to a confound of feature-engineering effort, tuning budget, or evaluation leniency that differs across papers reporting different hybrid configurations. A negative result under MBCA is a statement about *this specific, stated budget* at that insertion point, not a universal claim that no learned model could ever help there (Section 9 returns to this point).

A taxonomy of insertion points

Not every insertion point is equally exposed to failure. We classify each tested component by how much it can alter the trade set the rule-based system would otherwise produce:

- **Independent** — gates admissibility orthogonally to the entry signal; can only reject bars, never alter the signal itself. *Tested here:* Regime Filter.
- **Narrowing** — filters or re-weights a decision the rule-based system already made; cannot invent trades the rule-based system would not have taken. *Tested here:* Meta-Labeling, Position Sizing.
- **Replacing** — substitutes the rule-based mechanism outright; can generate a materially different trade set. *Tested here:* Entry, Exit.

We hypothesize, *ex ante*, that narrowing components bound their own worst-case damage (they can only subtract or re-weight, never add spurious decisions) and should therefore show gentler, lower-variance effects than replacing components, which can generate an entirely different trade set and therefore carry more variance in both directions. Section 6 tests this hypothesis directly; Figure 1 shows where each category sits in the pipeline.



MBCA discipline: swap exactly one insertion point at a time, everything else frozen

Held constant across every swap:
 8 identical backward-only features | identical LightGBM hyperparameters
 identical freeze → holdout → forward protocol | identical bootstrap-CI + deflated-Sharpe testing

The MBCA experiment: five insertion points into one rule-based pipeline, each with a matched-budget ML counterpart, and the invariants held constant across every swap.

The rule-based baseline (the MBCA control)

A JMA(7)/JMA(21) fast/slow crossover generates entries; exits are either a fixed $1.5 \times \text{ATR}$ stop / $3 \times \text{ATR}$ target bracket (forex) or a $1.5 \times \text{ATR}$ stop with a $2 \times \text{ATR}$ ratcheting trailing stop (commodities and Indian equities) — the exit mechanism each market’s own prior validation history in this repository selected before any ML work began. Position sizing is uniform (fixed-fractional risk). This baseline is our control: it is re-run inside every component’s own campaign script, so every comparison in this paper is against a freshly regenerated baseline under that exact campaign’s conditions.

Markets and data

- **Forex (4-pair):** AUDNZD, CADJPY, AUDUSD, NZDUSD, daily, already spans to present (no separate forward tail).
- **Commodities:** WTI, NATGAS, Brent, Gold, Silver, daily, with a separately-fetched forward tree.
- **Indian equities:** 48 NIFTY-50 constituents, daily, with annual rolling sector curation applied identically to every variant’s trade book, and a separately-fetched forward tree.

Crypto was not included in this campaign: daily crypto data exists in this repository but has no equivalent pre-ML validated baseline or forward-tail tree yet. We report this as a scope limitation, not a null result.

Components tested

Rule-based baseline; ML Regime Filter (*independent*; LightGBM trend/chop classifier gates entries, 3 label definitions); ML Meta-Labeling (*narrowing*; LightGBM classifier filters signal-fired trades by predicted $P(\text{win})$); ML Entry (*replacing*; two LightGBM classifiers generate entries directly, replacing `jma_signals`); ML Exit

(*replacing*; two LightGBM classifiers decide hold/exit per bar, replacing the ATR exit stack); ML Position Sizing (*narrowing*; the same frozen model as Meta-Labeling, consumed as a continuous capital multiplier instead of a threshold); and four combinations (Entry+Exit, Entry+Sizing, Exit+Sizing, Best Combined), detailed in Section 7.

Meta-Labeling and Position Sizing deliberately share one frozen model — identical features, label, and hyperparameters — differing only in whether its output is thresholded (accept/reject) or used as a continuous multiplier, isolating “threshold vs. scale” as the single manipulated variable between those two narrowing components.

Shared feature set and model

All components use the identical 8 backward-only features: ADX, ATR-normalized momentum, historical volatility, return z-score, backward-looking R^2 , RSI, MACD histogram, and KAMA slope — every one computable from data up to and including bar t , none from future bars. Every classifier is a `lightgbm.LGBMClassifier` with identical hyperparameters (`n_estimators=200`, `num_leaves=31`, `max_depth=6`, `learning_rate=0.05`, `min_child_samples=30`, `class_weight="balanced"`, fixed `random_state=42`) — no hyperparameter sweep for any ML component. This uniformity is the core MBCA constraint (Section 4.1): sweeping hyperparameters per component would confound component differences with tuning-effort differences, undermining the very attribution this paper sets out to make.

Freeze discipline

Every classifier is fit exactly once, on a pre-split slice of each market’s own data (a 70%-of-calendar-span split).

Training rows whose label lookahead window would cross the split date are excluded — enforced and unit-tested per component. After freezing, a classifier only ever predicts; none of the components refit on holdout or forward data. Enforced freeze discipline is essential to MBCA’s validity: any refit on evaluation data would confound result interpretation with dataset-selection bias.

Evaluation windows and statistics

Two windows per market where available: **holdout** (same processed tree the classifier trained on, **entry_time** $\geq$ split date) and **forward** (a genuinely separately-fetched tree the classifier never saw during training or holdout evaluation). Forex has no forward tree, reported as a limitation. Every comparison reports trade count, Sharpe ratio, max drawdown, deflated Sharpe ratio (**n_trials** set to the actual number of variants screened within that component), and a 95% bootstrap confidence interval of the difference in Sharpe between the ML variant and that same campaign’s own baseline. A result is significant iff this CI excludes zero.

Conventions, stated precisely. (i) A trade stream is converted to a daily return series by averaging the returns of all trades that *exit* on each date (realized-at-exit, not mark-to-market). (ii) Reported Sharpe ratios are annualized ($\sqrt{252}$) excess-return Sharpe ratios with a 6%/yr risk-free rate and population standard deviation. (iii) The bootstrap draws 2,000 replicates, resampling each variant’s daily series *independently* with replacement (i.i.d., unpaired, fixed seed), and computes the difference of *per-day, non-annualized* Sharpe ratios (mean/std); CI bounds are therefore on a scale $\sqrt{252} \approx 15.9$ times smaller than the annualized Δ Sharpe reported beside them. Earlier drafts described this test as a paired, instrument-level block bootstrap; that description was incorrect (see Appendix 13). Because the ML variant and its baseline trade the same instruments over the same days, their daily returns are typically positively correlated, and ignoring that pairing widens the interval (conservative); conversely, i.i.d. resampling ignores serial dependence, which can narrow it. The reference implementation exposes both the paper’s test and a date-paired alternative (`run_mbca(..., paired_ci=True)`).

Mathematical Formulation

For bar t in instrument i , $\mathbf{x}_{i,t} \in \mathbb{R}^8$ is the backward-only feature vector defined above, identical across all five components.

ML Regime Filter: $\hat{y}_{i,t}^{\text{regime}} = \mathbb{1}[P(\text{trend}_{i,t} = 1 \mid \mathbf{x}_{i,t}) \geq 0.5]$, where $\text{trend}_{i,t}$ is one of three self-labeled forward-window definitions.

ML Meta-Labeling: at each bar where the primary rule fires ($\text{signal}_{i,t} \neq 0$), let $r_{i,t}$ be the realized return of the trade the primary rule would open. The label is $y_{i,t}^{\text{meta}} = \mathbb{1}[r_{i,t} > 0]$; $\hat{p}_{i,t}^{\text{meta}} = P(y_{i,t}^{\text{meta}} = 1 \mid \mathbf{x}_{i,t})$ is estimated only at signal-fired bars, and the trade is taken iff $\hat{p}_{i,t}^{\text{meta}} \geq 0.5$.

ML Entry: two classifiers estimate, for direction $d \in \{+1, -1\}$, whether a hypothetical bracket entered at bar t (stop = $\text{close}_t - d \cdot 1.5 \cdot \text{ATR}_t$, target = $\text{close}_t + d \cdot 3.0 \cdot \text{ATR}_t$) hits target before stop within a 40-bar horizon: $\hat{p}_{d,i,t}^{\text{entry}} = P(\text{target-first} \mid \mathbf{x}_{i,t}, d)$. The realized signal is $\text{sign}(\arg\max_d \hat{p}_{d,i,t}^{\text{entry}})$ where that maximal probability also clears 0.5, else flat.

ML Exit: two classifiers estimate, for a position already open in direction d , whether price will still be favorable 10 bars ahead: $\hat{p}_{d,i,t}^{\text{exit}} = P(d \cdot (\text{close}_{t+10} - \text{close}_t) > 0 \mid \mathbf{x}_{i,t})$, evaluated at every bar while the position is open. Exit fires at the first bar $\hat{p}_{d,i,t}^{\text{exit}} < 0.5$, or at a $3 \times \text{ATR}$ safety-net stop, whichever comes first.

ML Position Sizing: reuses $\hat{p}_{i,t}^{\text{meta}}$, consumed as a continuous multiplier: $m_{i,t} = \text{clip}(2 \cdot \hat{p}_{i,t}^{\text{meta}}, 0, 2)$, and every trade’s realized return is scaled: $r'_{i,t} = m_{i,t} \cdot r_{i,t}$.

Generalization gap and significance: for metric M and component c , $\Delta M_c = M_{\text{ML}}(c) - M_{\text{baseline}}$, per market and window. Significance is assessed by an unpaired i.i.d. bootstrap of the two daily return series (2,000 replicates): with $\bar{S}R(r) = \bar{r}/s(r)$ the per-day Sharpe ratio and r^* a resample, the 95% CI is the 2.5th/97.5th percentile of $\bar{S}R(r_{\text{ML}}^*) - \bar{S}R(r_{\text{base}}^*)$ across replicates; a result is called significant iff the CI excludes zero.

Results: Component Attribution

Table 1 reports every component $\times$ market $\times$ window combination with a bootstrap CI available (40 rows, individual components only; combinations follow in Section 7). **0 of 40** configurations show a significant improvement; **10 of 40** show a significant loss. Figure 2 summarizes the same evidence, plus the four combinations, as the ML Opportunity Matrix; Figure 3 shows the full per-configuration forest plot.

The ML Opportunity Matrix: mean Δ Sharpe by component/combination and market
(cell text = mean Δ Sharpe across available windows; marker = cross-window consistency; row label color = component taxonomy)

	Forex	Commodities	Indian Equities	Overall Recommendation
Regime Filter	● -4.88	○ -1.67	○ -0.98	Avoid Significant loss in forex, commodities
Meta-Labeling	◇ -4.19	● +0.75	● +1.44	Mixed Avoid in forex (significant loss); promising in commodities, indian equities (unreplicated)
Entry	◇ -4.46	● -1.61	○ +0.36	Mixed Avoid in forex (significant loss); promising in indian equities (unreplicated)
Exit	◇ -6.48	● -4.84	○ -1.49	Avoid Significant loss in forex, commodities, indian equities
Position Sizing	◇ -1.03	○ -0.44	● +0.49	Worth exploring Worth exploring in indian equities (unreplicated); neutral/negative elsewhere
Entry + Exit	◇ -4.88	● -4.49	● -2.65	Avoid Significant loss in forex, commodities, indian equities
Entry + Sizing	◇ -3.99	● -2.44	○ +0.50	Mixed Avoid in forex (significant loss); promising in indian equities (unreplicated)
Exit + Sizing	◇ -6.21	● -4.43	○ -0.90	Avoid Significant loss in forex, commodities, indian equities
Best Combined	◇ -4.27	● +0.55	● +1.40	Mixed Avoid in forex (significant loss); promising in commodities, indian equities (unreplicated)

■ significant loss ■ non-sig. negative ■ ~zero ■ non-sig. positive ■ significant gain
● consistent sign across windows ○ inconsistent sign across windows ◇ single window only (e.g. forex)

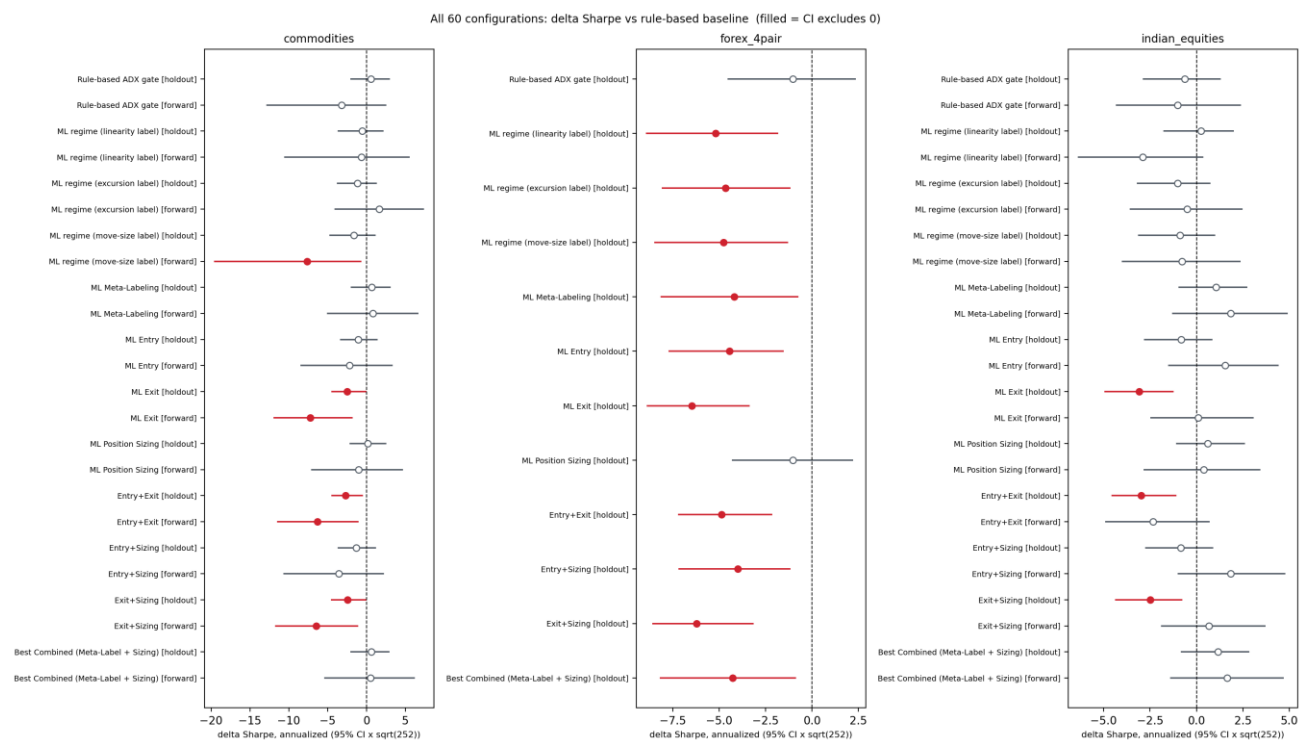
The ML Opportunity Matrix (flagship): every component and combination against every market, with a mechanically-derived Overall Recommendation. Cell color encodes effect direction and significance; marker encodes cross-window consistency; row-label color encodes the taxonomy of Section 4.2. Recommendations follow directly from significance, sign, and consistency (Section 4.1); none are hand-picked.

Individual-component attribution: every component $\times$ market $\times$ window with a bootstrap CI available (40 configurations).

Component	Variant	Market	Window	Δ Sharpe	Deflated Sharpe	Significant?
ML Entry	ML entry (replaces jma_signals)	commodities	forward	-2.17	0.857	no
ML Entry	ML entry (replaces jma_signals)	commodities	holdout	-1.06	0.986	no
ML Entry	ML entry (replaces jma_signals)	forex_4pair	holdout	-4.46	0.886	yes
ML Entry	ML entry (replaces jma_signals)	indian_equities	forward	+1.53	0.997	no
ML Entry	ML entry (replaces jma_signals)	indian_equities	holdout	-0.82	1.000	no
ML Exit	ML exit (replaces ATR stop/target)	commodities	forward	-7.22	0.031	yes
ML Exit	ML exit (replaces ATR stop/target)	commodities	holdout	-2.47	0.695	yes
ML Exit	ML exit (replaces ATR stop/target)	forex_4pair	holdout	-6.48	0.204	yes

Component	Variant	Market	Window	Δ Sharpe	Deflated Sharpe	Significant?
ML Exit	ML exit (replaces ATR stop/target)	indian_equities	forward	+0.09	0.946	no
ML Exit	ML exit (replaces ATR stop/target)	indian_equities	holdout	-3.08	0.444	yes
ML Meta-Labeling	ML meta-label filter	commodities	forward	+0.83	0.997	no
ML Meta-Labeling	ML meta-label filter	commodities	holdout	+0.67	1.000	no
ML Meta-Labeling	ML meta-label filter	forex_4pair	holdout	-4.19	0.854	yes
ML Meta-Labeling	ML meta-label filter	indian_equities	forward	+1.84	0.997	no
ML Meta-Labeling	ML meta-label filter	indian_equities	holdout	+1.04	1.000	no
ML Position Sizing	ML position sizing (re-weighted trades)	commodities	forward	-1.04	0.969	no
ML Position Sizing	ML position sizing (re-weighted trades)	commodities	holdout	+0.16	1.000	no
ML Position Sizing	ML position sizing (re-weighted trades)	forex_4pair	holdout	-1.03	1.000	no
ML Position Sizing	ML position sizing (re-weighted trades)	indian_equities	forward	+0.38	0.908	no
ML Position Sizing	ML position sizing (re-weighted trades)	indian_equities	holdout	+0.60	1.000	no
ML Regime Filter	Rule-based ADX gate	commodities	forward	-3.23	0.325	no
ML Regime Filter	ML regime (linearity label)	commodities	forward	-0.67	0.723	no
ML Regime Filter	ML regime (excursion label)	commodities	forward	+1.63	0.976	no
ML Regime Filter	ML regime (move-size label)	commodities	forward	-7.65	0.032	yes
ML Regime Filter	Rule-based ADX gate	commodities	holdout	+0.54	0.998	no
ML Regime Filter	ML regime (linearity label)	commodities	holdout	-0.54	0.885	no
ML Regime Filter	ML regime (excursion label)	commodities	holdout	-1.16	0.819	no
ML Regime Filter	ML regime (move-size label)	commodities	holdout	-1.65	0.545	no
ML Regime Filter	Rule-based ADX gate	forex_4pair	holdout	-1.02	0.997	no
ML Regime Filter	ML regime (linearity label)	forex_4pair	holdout	-5.20	0.313	yes
ML Regime Filter	ML regime (excursion label)	forex_4pair	holdout	-4.66	0.477	yes
ML Regime Filter	ML regime (move-size label)	forex_4pair	holdout	-4.78	0.455	yes
ML Regime Filter	Rule-based ADX gate	indian_equities	forward	-1.01	0.661	no
ML Regime Filter	ML regime (linearity label)	indian_equities	forward	-2.89	0.133	no
ML Regime	ML regime (excursion label)	indian_equities	forward	-0.51	0.895	no

Component	Variant	Market	Window	Δ Sharpe	Deflated Sharpe	Significant?
Filter						
ML Regime Filter	ML regime (move-size label)	indian_equities	forward	-0.78	0.787	no
ML Regime Filter	Rule-based ADX gate	indian_equities	holdout	-0.63	0.989	no
ML Regime Filter	ML regime (linearity label)	indian_equities	holdout	+0.23	1.000	no
ML Regime Filter	ML regime (excursion label)	indian_equities	holdout	-1.01	0.971	no
ML Regime Filter	ML regime (move-size label)	indian_equities	holdout	-0.89	0.957	no



All 60 configurations, forest plot with 95% bootstrap CIs, one panel per market. Dots are annualized Δ Sharpe; intervals are the per-day bootstrap CIs rescaled by $\sqrt{252}$ so both share one axis (earlier drafts drew the unscaled per-day interval around the annualized point; significance calls are unaffected). Filled = CI excludes zero.

Independent components: the regime filter

The regime filter is a dead end everywhere tested. All three ML label variants (linearity, excursion, move-size) lose against the rule-based ADX gate's own already-weak baseline in every market; one configuration (move-size label, commodities forward) loses significantly. **Why:** an independent gate can only reject bars; it cannot compensate for a weak or noisy label definition by learning from the entry signal itself, since it never sees it. The rule-based ADX gate this component is meant to improve on already underperforms not gating at all on these three markets (a pre-ML finding this repository established before any ML work began), so an ML variant of a rule that itself adds little has little to inherit.

Narrowing components: meta-labeling and position sizing

Meta-labeling and position sizing show the only encouraging (still non-significant) pattern in the entire campaign, concentrated in Indian equities: both beat baseline on holdout and forward Sharpe there (Meta-Labeling: Δ Sharpe +1.04 holdout, +1.84 forward; Position Sizing: +0.60 holdout, +0.38 forward). Position sizing — the identical underlying win-probability model, consumed as a continuous multiplier rather than a threshold — replicates the same directional pattern with a smaller effect size. **Why:** both components can only subtract from or re-weight the rule-based system's own trade set; they cannot invent new entries or exits, which bounds their downside relative to the replacing components below. Forex is again the exception: meta-labeling loses significantly there (the only significant loss either narrowing component produces).

Replacing components: entry and exit

Entry. ML Entry loses on forex (significantly), is mixed on commodities, and is marginally positive but non-significant on Indian equities. **Why:** entry replacement discards the rule-based JMA crossover's own trend-detection logic entirely and asks a classifier to rediscover it from the same 8 features with no lookahead advantage; on forex, where the rule-based baseline already achieves the campaign's highest Sharpe (5.53 on holdout), there is apparently little signal left for a learned entry model to find, and every departure from the rule looks like noise.

Exit. ML Exit is the clearest, most consistent failure mode of any single component: it loses significantly in 4 of 5 tested configurations. **Why, mechanistically, not just observationally:** trade count inflates $2.7\times-4.2\times$ across every market, because an exit that fires sooner than the rule-based ATR bracket un-blocks bars that a longer-held baseline trade would have absorbed, letting intervening JMA crossovers become their own separate, smaller, fee-eaten trades. This is a structural property of replacing a *holding* mechanism (a bracket that commits capital for a variable duration) with a per-bar *continuation classifier* that re-evaluates every bar independently — not a symptom of an undertrained or mistuned model.

Effect size scales with how much a component can alter the realized trade set, consistent with the Section 4.2 taxonomy's ex-ante hypothesis: position sizing (never rejects a trade) does the least damage; meta-labeling (rejects trades, keeps entry/exit fixed) is more

Combination experiments: 4 combinations $\times$ 3 markets $\times$ available windows (20 configurations).

volatile; entry and exit replacement (discard the rule-based mechanism outright) produce the largest swings, almost always negative.

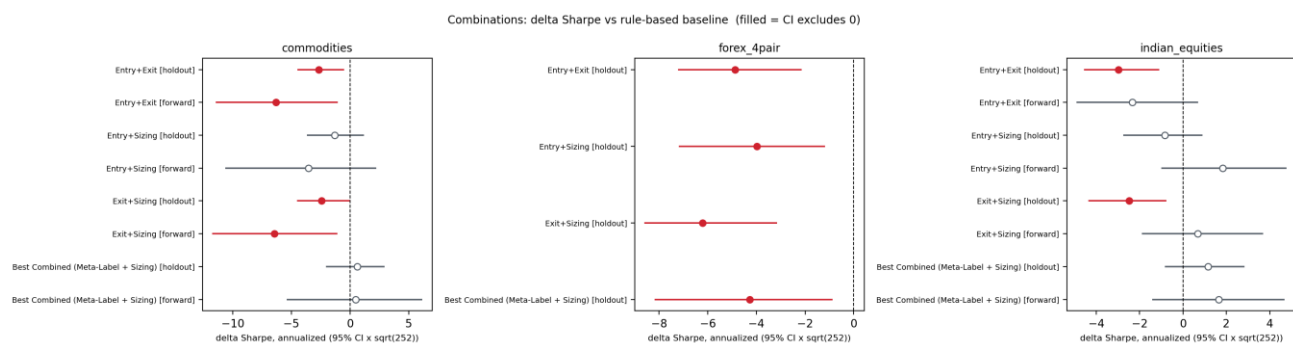
Statistical validation

Every Δ Sharpe carries a 95% bootstrap CI (2,000 replicates; Section 4), and every Sharpe is reported alongside its deflated Sharpe ratio, correcting for the actual number of variants screened within that component. No pooled cross-market or cross-component test is attempted: markets have too few instruments (4–48) and components too few evaluation windows for a meaningful meta-analytic pooling beyond the descriptive Opportunity Matrix itself; Section 9 discusses multiple-comparisons handling in more detail.

Combinations

Four combination experiments, run under the identical freeze/holdout/forward protocol and frozen-model-reuse discipline as every individual component: (1) **Entry+Exit** — ML Entry's signal generation feeding ML Exit's continuation-based exit mechanism; (2) **Entry+Sizing** — ML Entry generating trades, ML Position Sizing re-weighting them; (3) **Exit+Sizing** — rule-based entry, ML Exit's exit mechanism, ML Position Sizing re-weighting; (4) **Best Combined** — chosen empirically, not assumed in advance: ML Meta-Labeling composed with ML Position Sizing, the two *narrowing* components that individually showed the paper's only encouraging pattern (Section 4.2).

Market	Window	Variant	Trades	Sharpe (base $\rightarrow$ variant)	Deflated Sharpe	Δ Sharpe	Significant?
commodities	forward	best_combined	37	4.17 $\rightarrow$ 4.67	0.997	+0.49	no
commodities	forward	entry_exit	210	4.17 $\rightarrow$ -2.14	0.106	-6.31	yes
commodities	forward	entry_sizing	65	4.17 $\rightarrow$ 0.62	0.619	-3.55	no
commodities	forward	exit_sizing	113	4.17 $\rightarrow$ -2.28	0.123	-6.45	yes
commodities	holdout	best_combined	170	2.72 $\rightarrow$ 3.34	1.000	+0.61	no
commodities	holdout	entry_exit	1420	2.72 $\rightarrow$ 0.05	0.618	-2.68	yes
commodities	holdout	entry_sizing	335	2.72 $\rightarrow$ 1.40	0.957	-1.33	no
commodities	holdout	exit_sizing	610	2.72 $\rightarrow$ 0.30	0.723	-2.42	yes
forex_4pair	holdout	best_combined	118	5.53 $\rightarrow$ 1.26	0.831	-4.27	yes
forex_4pair	holdout	entry_exit	1233	5.53 $\rightarrow$ 0.66	0.977	-4.88	yes
forex_4pair	holdout	entry_sizing	243	5.53 $\rightarrow$ 1.54	0.966	-3.99	yes
forex_4pair	holdout	exit_sizing	596	5.53 $\rightarrow$ -0.68	0.321	-6.21	yes
indian_equities	forward	best_combined	197	1.05 $\rightarrow$ 2.70	0.994	+1.65	no
indian_equities	forward	entry_exit	1794	1.05 $\rightarrow$ -1.29	0.089	-2.34	no
indian_equities	forward	entry_sizing	290	1.05 $\rightarrow$ 2.89	0.999	+1.83	no
indian_equities	forward	exit_sizing	915	1.05 $\rightarrow$ 1.73	0.991	+0.68	no
indian_equities	holdout	best_combined	527	2.84 $\rightarrow$ 4.01	1.000	+1.16	no
indian_equities	holdout	entry_exit	3420	2.84 $\rightarrow$ -0.13	0.586	-2.97	yes
indian_equities	holdout	entry_sizing	756	2.84 $\rightarrow$ 2.00	1.000	-0.84	no
indian_equities	holdout	exit_sizing	1279	2.84 $\rightarrow$ 0.36	0.773	-2.48	yes



The 20 combination-only rows, same convention as Figure 3.

Any combination containing ML Exit inherits its overtrading failure, and composition does not average it away. Entry+Exit and Exit+Sizing both explode trade count and both lose, significantly on 6 of 8 testable holdout/forward pairs. Composing ML Exit with a component that individually looked fine (Position Sizing) does not rescue it — consistent with the turnover-inflation mechanism identified above being a property of the exit mechanism itself, not something a downstream narrowing component can correct after the fact.

Entry+Sizing is gentler than ML Entry alone, consistent with a dose-response reading: position sizing can only re-weight the trades ML Entry generates, not reject them, so it softens but does not reverse ML Entry's forex loss, and turns indian_equities forward strongly positive (1.05 → 2.89), though the CI still touches zero.

Best Combined — composing two narrowing components — is the campaign's standout result, and still falls short of significance. Indian-equities holdout Sharpe improves from 2.84 to 4.01 with a bootstrap CI of $[-0.050, 0.176]$, the closest any of the 60 tested configurations comes to excluding zero. Forex remains the exception: Best Combined still loses there significantly, reinforcing that forex is hostile to the mechanism itself, not merely to any one component's implementation.

Composition amplifies a shared directional pattern rather than creating value the individual components lacked. Best Combined's indian_equities holdout Sharpe (4.01) exceeds both Meta-Labeling alone (3.89) and Position Sizing alone (3.44), suggesting the two narrowing components' gains are at least partially additive here — consistent with the taxonomy prediction that narrowing components address different axes of the same decision (which trades to keep, vs. how much to risk on them) and so do not simply duplicate each other's effect. This is, however, a single freeze-date observation (Section 9).

Discussion

What MBCA reveals

Across 60 tested configurations, zero showed a statistically significant improvement over the rule-based baseline, even when the two most promising components were composed together. Read at face value, this could

suggest “ML does not help hybrid trading systems.” MBCA's contribution is to show that this framing is both too strong and not useful: the honest reading is narrower and more actionable. ML's contribution here is component-specific and market-specific, and the Section 4.2 taxonomy predicts much of that heterogeneity ex ante: narrowing components (which can only subtract or re-weight) are gentler and occasionally directionally positive; replacing components (which can generate an entirely different trade set) are more volatile and, for exit specifically, actively harmful via an identified turnover mechanism that propagates into any combination containing it; and the one independent component tested, the regime filter, is a dead end everywhere, inheriting the weakness of the rule-based gate it was meant to improve on.

Why forex is different

Forex is hostile to every ML intervention and every combination tested: every configuration on that sleeve loses on holdout, and most losses are statistically significant. One plausible explanation is that the rule-based 4-pair JMA+ATR recipe already extracts most of the value available in this feature set on this sleeve — its own rule-based holdout Sharpe (5.53) is the highest of any market's baseline in this study — leaving little residual signal for a learned model, at this feature budget, to find. Every departure from the rule, in this reading, looks like noise rather than an improvement. This is a hypothesis consistent with the pattern, not something we can prove definitively from a single freeze date; the alternative explanation — that this particular feature set or model class is simply mismatched to forex specifically — cannot be ruled out from this evidence alone (Section 9). The independent replication (Section 11) weakens this finding further: with different forex data and a flat cost model, the rule-based baseline is much weaker and only 2 of 12 forex configurations lose significantly, so “forex is hostile to ML” should be read as specific to the original data and cost setup.

Why composition amplifies rather than creates

Composing two individually-non-negative components (Best Combined) amplifies their shared directional pattern — Indian-equities holdout Sharpe reaches 4.01, above either component alone — rather than cancelling it. But amplification of a non-significant effect is still non-significant. This is a useful negative result in its own right: it means the near-miss is not an artifact of noise in one

component being fortuitously offset by noise in another (which would predict regression toward zero on composition), but a case where two components pointing the same direction reinforce each other. That reinforcement pattern is itself evidence the underlying signal, while too small to call significant at this sample size, is not pure noise — which is precisely why we flag Best Combined, and not any other configuration, as the single one worth a second-freeze-date replication.

Toward MBCA beyond trading

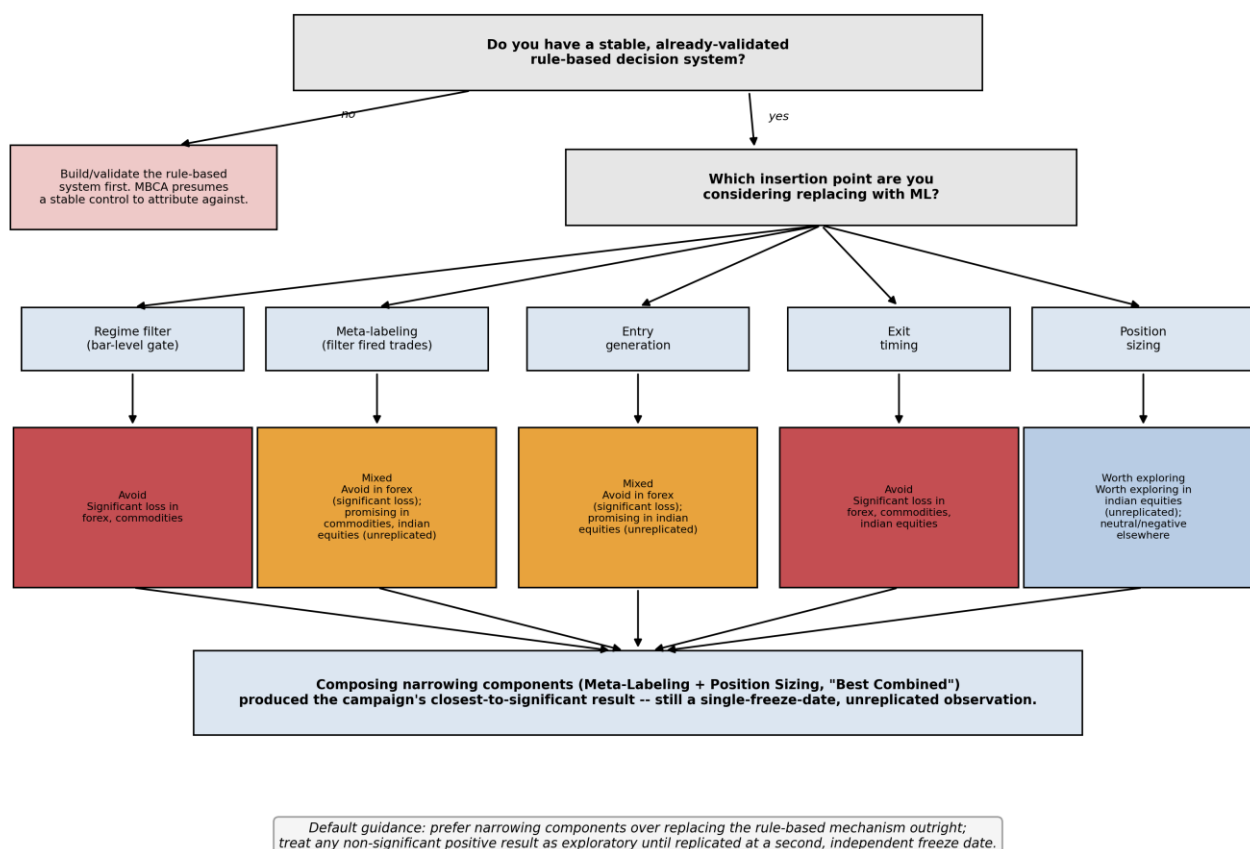
We designed and applied MBCA to one trading strategy because it gave us an already-validated rule-based control and a market structure (three distinct markets) that let us test whether component-level patterns generalize across data-generating processes without changing the pipeline. Nothing in MBCA's definition (Section 4.1) is trading-specific: it applies wherever a system combines a rule-based or otherwise validated control with candidate learned components at distinct decision points — for instance, control systems that combine PID controllers with learned correction terms, recommendation pipelines that combine heuristic ranking with learned re-ranking stages, or clinical decision-support tools that combine

guideline-based triage with learned risk scores. We propose this as a direction, not a demonstrated result: this paper tests MBCA on one domain, and we have not run it on a non-trading system. Section 10 lists this as the most valuable extension for establishing MBCA's generality beyond the claim we can support here.

Practical implications

For practitioners with an already-validated rule-based system, this evidence (Figure 2, Figure 5) suggests: skip an independent regime-filter gate unless the underlying rule-based gate itself has separately demonstrated value; prioritize narrowing components (meta-labeling, position sizing) over replacing components when adding ML, particularly in markets showing structural regime shifts (Indian equities, in this study); avoid learned entry/exit replacement, since the rule-based mechanism appears to already be near-optimal, or the replacement introduces new failure modes (turnover inflation), given this feature/model budget; and treat any positive, non-significant result — including Best Combined — as exploratory until replicated at a second, independent freeze date.

When should ML replace a rule-based component? A decision guide grounded in MBCA evidence



A decision guide for whether to convert a given insertion point to ML, grounded directly in the Section 6–7 evidence; every leaf reuses the same mechanically-derived recommendation shown in Figure 2.

Limitations

We address five objections directly, in order of severity.

- **Every result is a single freeze-date observation per market.** This is the paper’s most serious limitation. No component/market pairing — individual or combined — has been replicated at a second, independent split point. It applies with particular force to Best Combined’s near-miss result in Indian equities (bootstrap CI $[-0.050, 0.176]$): a single-freeze-date confidence interval that happens to sit close to excluding zero is not evidence of a real effect on its own, and we do not treat it as one. We report it as the single configuration most worth a second-freeze-date replication (Section 10), not as a validated finding.
- **One feature set, one model class.** This is deliberate, and it is the point of MBCA, not an oversight: sweeping features or hyperparameters per component would confound component differences with tuning-effort differences, which is exactly the confound MBCA is designed to eliminate (Section 4.1). But this means every negative result in this paper is a statement about *this specific budget* at that insertion point — 8 backward-only features, one LightGBM configuration — not a universal claim that no learned policy could ever add value there. A different feature set (e.g. market microstructure, alternative data) or model class might change individual results; MBCA’s contribution is the protocol for testing that fairly, not a claim that this paper has exhausted the space of possible ML configurations at each insertion point.
- **No calibration analysis.** We report trading-performance metrics (Sharpe, CAGR, drawdown, deflated Sharpe, bootstrap CI) throughout; we do not report classifier calibration metrics (Brier score, expected calibration error, reliability diagrams) for any component’s underlying classifier. This is a scope decision, not an oversight, and we do not claim otherwise: a well-calibrated classifier can still produce a poorly-performing trading component (if its features carry little signal), and a poorly-calibrated one can still trade profitably (if its rank-ordering is useful even when its probabilities are not). Calibration analysis is a natural, separate extension (Section 10), not a claim we make here.
- **Multiple-comparisons handling.** Each Sharpe is reported alongside a deflated Sharpe ratio that corrects for the number of variants screened *within that component’s own campaign* (e.g. `n_trials=3` for the Regime Filter’s three label variants). We do not apply a single global correction (e.g. Bonferroni) across all 60

tested configurations, because the paper’s claim is component-specific and market-specific, not a claim that ML generally helps trading. A reader who prefers a fully conservative, globally-corrected reading should treat every non-significant result in this paper (including Best Combined) as, if anything, more exploratory still — our own framing already treats it that way.

- **Matched-budget fairness.** A natural objection: forcing every component to share identical hyperparameters might handicap components that would benefit from different tuning. We accept this trade-off explicitly, because the alternative — tuning each component separately — answers a different question (“which component performs best given unlimited tuning effort”) than the one MBCA is designed to answer (“holding effort fixed, where does ML add value”). Per-component hyperparameter sensitivity is listed as future work (Section 10), not folded into this paper’s headline claims.

Two further, narrower limitations carry over from the underlying strategy validation:

- **Forex has no forward tree** — holdout-only evidence for that market.
- **Crypto was not included** — no equivalent pre-ML validated baseline or forward-tail tree exists yet for that market; this is a scope limitation, not a null result.
- **No claim of a validated, deployable trading strategy.** Nothing here is paper-traded, and no positive trading-performance claim is made.

Future Work: Extending MBCA

- **Second, independent freeze-date replication** per market, prioritizing Best Combined (Meta-Labeling + Position Sizing) in Indian equities first — the single most valuable next step, since it is the one configuration whose current evidence is closest to informative.
- **Extend the component-attribution protocol to crypto** once an equivalent pre-ML validated baseline and forward-tail tree exist.
- **Test MBCA on a non-trading hybrid system** (e.g. a rule-based control loop with a learned correction term, or a heuristic ranking pipeline with a learned re-ranking stage) to establish whether the taxonomy and matrix format proposed in Section 8 generalize beyond trading, which this paper does not itself demonstrate.
- **Classifier calibration analysis** (Brier score, reliability diagrams) per component, to test whether calibration quality predicts which components are worth converting to ML independent of realized trading performance.

- **Alternative budgets:** investigate whether a minimum-hold-bars floor or a K-consecutive-signal exit rule removes ML Exit’s turnover-inflation failure mode, and whether this rescues Entry+Exit/Exit+Sizing; and whether a different feature set or model class changes which components fall into which taxonomy bucket.
- **Shared model refit:** test the Best Combined pairing against a shared, jointly-fit meta-label model rather than the campaign’s current independent refit inside `PositionSizeModel`, to rule out any avoidable variance in the near-miss result.

Reference Implementation and Independent Replication

We release `mbca`, a self-contained Python package (pandas, scikit-learn, LightGBM) that implements MBCA end to end: the eight shared features, the rule-based pipeline and walk-forward harness, the five components as drop-in replacements of one pipeline function each, the

Paper vs. independent replication (`mbca`), Δ Sharpe (annualized) per configuration.

freeze protocol, and the statistics of Section 4. Its test suite enforces the paper’s methodological invariants directly: features computed on data truncated at bar t equal those computed on the full series (no lookahead); scrambling every post-split bar leaves every fitted model bit-for-bit unchanged (freeze); position sizing never adds or removes a trade; and components compose by replacing exactly one insertion point. A user supplies their own entry rule, exit rule, and data, and obtains the full attribution table and Opportunity Matrix for their strategy.

Because `mbca` was written from scratch rather than copied from the study pipeline, re-running the full 60-configuration grid with it is an independent replication, with three known differences: the forex data come from a different vendor with a flat 7 bps cost instead of per-pair spread and swap; Indian equities are traded without the annual sector-curation overlay; and one walk-forward over the full history replaces the separately-fetched forward tree. Table 3 compares every configuration.

Market	Window	Variant	Paper	Replication	Paper verdict	Replication verdict
commodities	forward	Rule-based ADX gate	-3.23	-5.03	n.s.	n.s.
commodities	forward	ML Regime Filter (linearity)	-0.67	-3.51	n.s.	n.s.
commodities	forward	ML Regime Filter (excursion)	+1.63	+2.13	n.s.	n.s.
commodities	forward	ML Regime Filter (move_size)	-7.65	+1.63	sig. loss	n.s.
commodities	holdout	Rule-based ADX gate	+0.54	+0.51	n.s.	n.s.
commodities	holdout	ML Regime Filter (linearity)	-0.54	+2.19	n.s.	n.s.
commodities	holdout	ML Regime Filter (excursion)	-1.16	+0.20	n.s.	n.s.
commodities	holdout	ML Regime Filter (move_size)	-1.65	-0.81	n.s.	n.s.
forex	holdout	Rule-based ADX gate	-1.02	-1.05	n.s.	n.s.
forex	holdout	ML Regime Filter (linearity)	-5.20	-0.02	sig. loss	n.s.
forex	holdout	ML Regime Filter (excursion)	-4.66	-3.55	sig. loss	n.s.
forex	holdout	ML Regime Filter (move_size)	-4.78	-2.16	sig. loss	n.s.
indian_equities	forward	Rule-based ADX gate	-1.01	-0.43	n.s.	n.s.
indian_equities	forward	ML Regime Filter (linearity)	-2.89	-0.21	n.s.	n.s.
indian_equities	forward	ML Regime Filter (excursion)	-0.51	-0.43	n.s.	n.s.
indian_equities	forward	ML Regime Filter (move_size)	-0.78	+0.29	n.s.	n.s.
indian_equities	holdout	Rule-based ADX gate	-0.63	-1.39	n.s.	n.s.
indian_equities	holdout	ML Regime Filter (linearity)	+0.23	-1.35	n.s.	n.s.
indian_equities	holdout	ML Regime Filter (excursion)	-1.01	-1.27	n.s.	n.s.
indian_equities	holdout	ML Regime Filter (move_size)	-0.89	-1.69	n.s.	sig. loss
commodities	forward	ML Meta-Labeling	+0.83	-0.47	n.s.	n.s.
commodities	holdout	ML Meta-Labeling	+0.67	+1.27	n.s.	n.s.
forex	holdout	ML Meta-Labeling	-4.19	+0.94	sig. loss	n.s.
indian_equities	forward	ML Meta-Labeling	+1.84	+1.40	n.s.	n.s.
indian_equities	holdout	ML Meta-Labeling	+1.04	-0.02	n.s.	n.s.
commodities	forward	ML Entry	-2.17	-3.50	n.s.	n.s.
commodities	holdout	ML Entry	-1.06	-0.12	n.s.	n.s.

Market	Window	Variant	Paper	Replication	Paper verdict	Replication verdict
forex	holdout	ML Entry	-4.46	-1.06	sig. loss	n.s.
indian_equities	forward	ML Entry	+1.53	+0.37	n.s.	n.s.
indian_equities	holdout	ML Entry	-0.82	-1.86	n.s.	sig. loss
commodities	forward	ML Exit	-7.22	-6.16	sig. loss	sig. loss
commodities	holdout	ML Exit	-2.47	-1.61	sig. loss	n.s.
forex	holdout	ML Exit	-6.48	-3.87	sig. loss	sig. loss
indian_equities	forward	ML Exit	+0.09	-0.12	n.s.	n.s.
indian_equities	holdout	ML Exit	-3.08	-5.07	sig. loss	sig. loss
commodities	forward	ML Position Sizing	-1.04	-1.13	n.s.	n.s.
commodities	holdout	ML Position Sizing	+0.16	+0.18	n.s.	n.s.
forex	holdout	ML Position Sizing	-1.03	-0.79	n.s.	n.s.
indian_equities	forward	ML Position Sizing	+0.38	+0.18	n.s.	n.s.
indian_equities	holdout	ML Position Sizing	+0.60	+0.16	n.s.	n.s.
commodities	forward	Entry+Exit	-6.31	-7.19	sig. loss	sig. loss
commodities	forward	Entry+Sizing	-3.55	-5.86	n.s.	sig. loss
commodities	forward	Exit+Sizing	-6.45	-5.45	sig. loss	sig. loss
commodities	forward	Best Combined (Meta-Label + Sizing)	+0.49	-0.93	n.s.	n.s.
commodities	holdout	Entry+Exit	-2.68	-1.66	sig. loss	n.s.
commodities	holdout	Entry+Sizing	-1.33	-0.34	n.s.	n.s.
commodities	holdout	Exit+Sizing	-2.42	-1.31	sig. loss	n.s.
commodities	holdout	Best Combined (Meta-Label + Sizing)	+0.61	+1.19	n.s.	n.s.
forex	holdout	Entry+Exit	-4.88	-3.34	sig. loss	n.s.
forex	holdout	Entry+Sizing	-3.99	-0.93	sig. loss	n.s.
forex	holdout	Exit+Sizing	-6.21	-3.91	sig. loss	sig. loss
forex	holdout	Best Combined (Meta-Label + Sizing)	-4.27	+1.11	sig. loss	n.s.
indian_equities	forward	Entry+Exit	-2.34	-2.44	n.s.	n.s.
indian_equities	forward	Entry+Sizing	+1.83	+0.36	n.s.	n.s.
indian_equities	forward	Exit+Sizing	+0.68	+0.40	n.s.	n.s.
indian_equities	forward	Best Combined (Meta-Label + Sizing)	+1.65	+1.53	n.s.	n.s.
indian_equities	holdout	Entry+Exit	-2.97	-5.02	sig. loss	sig. loss
indian_equities	holdout	Entry+Sizing	-0.84	-2.13	n.s.	sig. loss
indian_equities	holdout	Exit+Sizing	-2.48	-4.39	sig. loss	sig. loss
indian_equities	holdout	Best Combined (Meta-Label + Sizing)	+1.16	+0.04	n.s.	n.s.

What replicates. (i) Zero of 60 configurations show a significant improvement, in the paper and in the replication. (ii) 49 of 60 configurations agree in the sign of Δ Sharpe. (iii) ML Exit is the most reliably harmful component: significant losses in all three markets, with trade count inflated 2.4–3.0 $\times$ (paper: 2.7–4.2 $\times$) — the turnover mechanism of Section 6 reappears in an independent implementation. (iv) Every combination containing ML Exit loses in 9 of 10 holdout/forward configurations in both runs. (v) The narrowing components remain the only ones directionally positive in more than one market.

What does not. (i) *Forex hostility is not robust.* The paper found significant losses in 10 of 12 forex configurations; the replication finds 2 of 12 (ML Exit and Exit+Sizing), and Meta-Labeling flips from -4.19 to $+0.94$. The rule-based forex baseline is also far weaker (holdout Sharpe 1.39 vs. 5.53), consistent with Section 8's hypothesis that ML looked hostile on forex because the baseline left little residual signal — a hypothesis that depends on the data vendor and cost model, not only on the market. (ii) *Best Combined's near-miss is only half replicated.* Its Indian-equities forward improvement reproduces closely ($+1.53$ vs. $+1.65$, CI $[-0.051, 0.251]$), but the holdout improvement does not ($+0.04$ vs. $+1.16$) once the sector-curation overlay is removed and the baseline itself rises to 4.31. (iii) The replication finds 12 significant losses rather than 20, mostly because of the forex difference.

Reading. The conclusions this paper draws at the level of the *method* and the *taxonomy* — no ML component significantly improves a validated rule-based system at this budget; replacing a holding mechanism with a per-bar continuation classifier is structurally harmful; narrowing components are the only candidates worth further study — survive an independent re-implementation. Conclusions at the level of a specific market or a specific configuration's effect size do not survive reliably, and we have weakened the corresponding claims above accordingly. This is the discipline MBCA recommends to its users: treat per-cell effects as exploratory until replicated.

Conclusion

We introduced Matched-Budget Component Attribution (MBCA), a controlled-variable methodology for deciding where, specifically, a machine-learned component adds value inside an already-validated hybrid decision system. MBCA holds data, features, model class, freeze discipline, and evaluation protocol fixed and varies only the insertion point, so that a resulting performance delta can be attributed to that insertion point rather than to a confound of tuning or feature-engineering effort.

Applying MBCA to one rule-based trading system across three markets and 60 configurations (5 components, 4 combinations, up to 2 evaluation windows each), we found zero statistically significant improvements and twenty statistically significant losses, concentrated in the forex sleeve and the ML Exit component (a turnover-

inflation failure mode identified and explained, not just observed, and inherited rather than averaged away by any combination containing it). A taxonomy of insertion points — independent, narrowing, replacing — predicted much of this heterogeneity *ex ante*: narrowing components (meta-labeling, position sizing) showed a mild, market-concentrated, non-significant positive pattern, while replacing components (entry, exit) were more volatile and often actively harmful. The one genuinely encouraging signal, Best Combined (Meta-Labeling filtering trades, then Position Sizing re-weighting the survivors), came closer to statistical significance in Indian equities (bootstrap CI $[-0.050, 0.176]$) than any other configuration tested, beat baseline on both windows in both commodities and Indian equities, and is the paper's single highest-priority target for a second-freeze-date replication.

We package this evidence as the ML Opportunity Matrix (Figure 2) and an accompanying decision flowchart (Figure 5) — reusable, mechanically-derived decision-support artifacts, not hand-picked recommendations. The paper's contribution is MBCA itself: a named, reusable methodology for holding everything constant, swapping or composing insertion points, and reporting every resulting configuration including the failures — offered as a protocol future work can apply to other hybrid systems, in trading and, we hypothesize but have not yet tested, beyond it.

Errata relative to Draft v3

No reported number changed. (1) The significance test was previously described as a paired, instrument-level block bootstrap; the code that produced every CI resamples each daily series independently and i.i.d. and compares per-day (non-annualized) Sharpe ratios. Section 4 now describes the test actually run and gives both scales. (2) Figures 3 and 4 previously drew the per-day CI around the annualized point estimate; they are regenerated on a common annualized axis. Significance calls are unchanged. (3) CAGR is no longer cited: on a realized-at-exit daily series it is not a capital-allocation growth rate. (4) Section 11 (independent replication) is new, and the forex and Best Combined claims have been qualified in its light.

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